C. Compliance Checking

Part 3 combines the Process Record and Rule Record using the rule–process matching and three violation definitions of Sun et al. [9]. Our contribution concerns Part 2; the checking formulas are retained below.

For rule r, A_r, P_r, and U_r denote its actions, actors, and sequential constraints. For process model m, A_m, E_m, and R_m denote activities, events, and actors. The functions $action$ and bs_obj extract actions and business objects from activity and event labels. The mappings f_r and f_m associate actors or business objects with their corresponding actions, and F_m denotes the process control-flow relation.

We compute text similarity $sim(x, y)$ using cosine similarity between MPNet-based Sentence-BERT embeddings [17], [18]. Thresholds τ, γ, and ϑ control rule–process matching, action matching, and actor comparison, respectively; ϑ corresponds to θ in the experimental results.

Rule–Process Matching. The set $D_{r,m}$ pairs each rule action with its most similar element in $action(A_m \cup E_m)$. Similarly, $O_{r,m}$ pairs each rule actor with its most similar element in $bs_obj(A_m \cup E_m) \cup R_m$. Sun's matching score is

$$\mathcal{F}(r, m, \tau) = \max\left(\frac{\sum_{(a_r, a_m) \in D_{r,m}} sim(a_r, a_m) > \tau}{|D_{r,m}|}, \frac{\sum_{(o_r, o_m) \in O_{r,m}} sim(o_r, o_m) > \tau}{|O_{r,m}|} \right). \tag{1}$$

The sums count mapping pairs whose similarity exceeds τ.

Missing-Action Violation. The score counts mapped rule actions with similarity below γ:

$$action_violation(r, m, \gamma) = \frac{\left|\{(a_r, a_m) \in D_{r,m} \mid sim(a_r, a_m) < \gamma\}\right|}{|A_r|}. \tag{2}$$

Incorrect-Actor Violation. Rule actors with matched actions form $R_{r,m,\gamma}$, and the corresponding shared process-side candidates form $C_{r,m,\gamma}$:

$$R_{r,m,\gamma} = \big\{ g \in P_r \mid \exists (a_r, a_m) \in D_{r,m}, \; f_r(g) = a_r \wedge sim(a_r, a_m) > \gamma \big\}, \tag{3}$$

$$C_{r,m,\gamma} = \big\{ r \in bs_obj(A_m \cup E_m) \cup R_m \mid \exists g \in R_{r,m,\gamma}, \; sim(f_r(g), f_m(r)) > \gamma \big\}. \tag{4}$$

A rule actor contributes to the score if a candidate in the shared set has similarity below ϑ:

$$actor_violation(r, m, \gamma, \vartheta) = \frac{\left|\{r \in R_{r,m,\gamma} \mid \exists r' \in C_{r,m,\gamma}, sim(r, r') < \vartheta\}\right|}{|R_{r,m,\gamma}|}. \tag{5}$$

Out-of-Order Violation. Sequential constraints with both endpoint actions matched form

$$U_{r,m,\gamma} = \big\{ (u_r, u'_r) \in U_r \mid \exists \big\{ (u_r, u_m) \in D_{r,m}, (u'_r, u'_m) \in D_{r,m} \big\}, \; sim(u_r, u_m) > \gamma \wedge sim(u'_r, u'_m) > \gamma \big\}. \tag{6}$$

The score is the fraction of evaluable constraints that do not satisfy the prescribed order:

$$order_violation(r, m, \gamma) = \frac{|U_{r,m,\gamma}| - \left|\big\{ (u_r, u'_r) \in U_{r,m,\gamma} \mid \exists \big\{ (u_r, u_m) \in D_{r,m}, (u'_r, u'_m) \in D_{r,m} \big\}, \; (u_m, u'_m) \in F_m \wedge (u'_m, u_m) \notin F_m \big\}\right|}{|U_{r,m,\gamma}|}. \tag{7}$$